

CSE3068 SEQUENTIAL AND SPATIAL DATA MINING

PROJECT REVIEW – I

RescuePath AI

Intelligent Disaster Response and Evacuation Platform

using Sequential and Spatial Data Mining

SUBMITTED BY

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1. PROBLEM STATEMENT

India is one of the world's most disaster-prone nations. Floods alone affect 7.5 million hectares of land annually and cause around 1,600 deaths every year (NDMA, 2024). States like Kerala, Assam, and Bihar face recurring, life-threatening flood events. Despite modern weather forecasting, existing disaster response systems operate in silos — rainfall is monitored by meteorological departments, road conditions by transport authorities, and evacuation decisions are made manually by officials with outdated information. There is no unified platform that predicts risk, maps danger zones, and recommends evacuation routes in real time.

Given historical and real-time disaster data, rainfall sequences, and road-network information, can we build a system that: (a) predicts flood risk for the next 24–72 hours using sequential data mining, (b) identifies spatially clustered high-risk zones using spatial data mining, and (c) recommends optimal evacuation routes avoiding danger areas — all within one integrated platform?

Key challenges being addressed:

- No integrated platform combining prediction, risk mapping, and routing in one place.
- No real-time automated risk zone mapping — current maps are static and outdated.
- Manual, slow evacuation planning under crisis conditions with no route optimization.
- Flood alerts raised only after flooding begins — no 24–72 hour advance prediction.
- Citizens have no accessible tool to check their area's risk or find a safe route.

2. PROGRAMMING LANGUAGE USED

Language	Version	Used For	Why Chosen
Python	3.11+	Back-end, ML models, GIS, APIs	Largest data science ecosystem; seamless library integration
JavaScript	ES2022	Front-end dashboard (React + Leaflet)	Required for interactive browser-based maps
SQL	Standard	Database queries (PostgreSQL+PostGIS)	Standard for relational and spatial data querying

3. TOOLS

Category	Tool / Library	Purpose
Data Processing	Pandas	Tabular data loading, cleaning, and manipulation
Data Processing	NumPy	Numerical array operations across all modules
Data Processing	Jupyter Notebook	Exploratory data analysis and model prototyping
Geospatial / GIS	GeoPandas	Spatial data frames, shapefiles, district-level mapping
Geospatial / GIS	OSMnx	Download OpenStreetMap road network as a graph

Geospatial / GIS	Rasterio	Read satellite raster imagery (GeoTIFF / NDVI)
Geospatial / GIS	Shapely + Fiona	Geometric object creation and GIS file format I/O
Geospatial / GIS	PyProj	Coordinate reference system transformations
Geospatial / GIS	PostGIS	Spatial SQL queries inside PostgreSQL database
Machine Learning	Scikit-learn	DBSCAN clustering, KDE, model evaluation metrics
Deep Learning	TensorFlow / Keras	LSTM model for 24–72 hour flood risk prediction
Routing	NetworkX	Road graph creation; Dijkstra/A* evacuation routing
Visualisation	Folium	Interactive HTML flood risk and route maps in Python
Visualisation	Matplotlib / Plotly	Static and interactive charts, training curves
Visualisation	Leaflet.js	Live interactive map rendering in the browser
Front-End	React.js	Component-based UI for the web dashboard
Back-End API	FastAPI + Uvicorn	REST API for prediction, risk zones, and routing
Database	PostgreSQL	Stores flood records, geometries, rainfall sequences
Database	SQLAlchemy	Python ORM for database interaction
Dev Tools	Git + GitHub	Version control and code collaboration
Dev Tools	VS Code / QGIS	Code editor and desktop GIS for dataset validation